Exp. No: 4

Analysis of Continuous Charge Distribution in electric field. To find the

(i) line charge density,

(ii) Surface charge density and

(iii) Volume charge density for any specific

(a) line,

(b) surface or

(c) volume.

Aim: To write program for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB.

Software required:

•SCILAB version 6.0.1

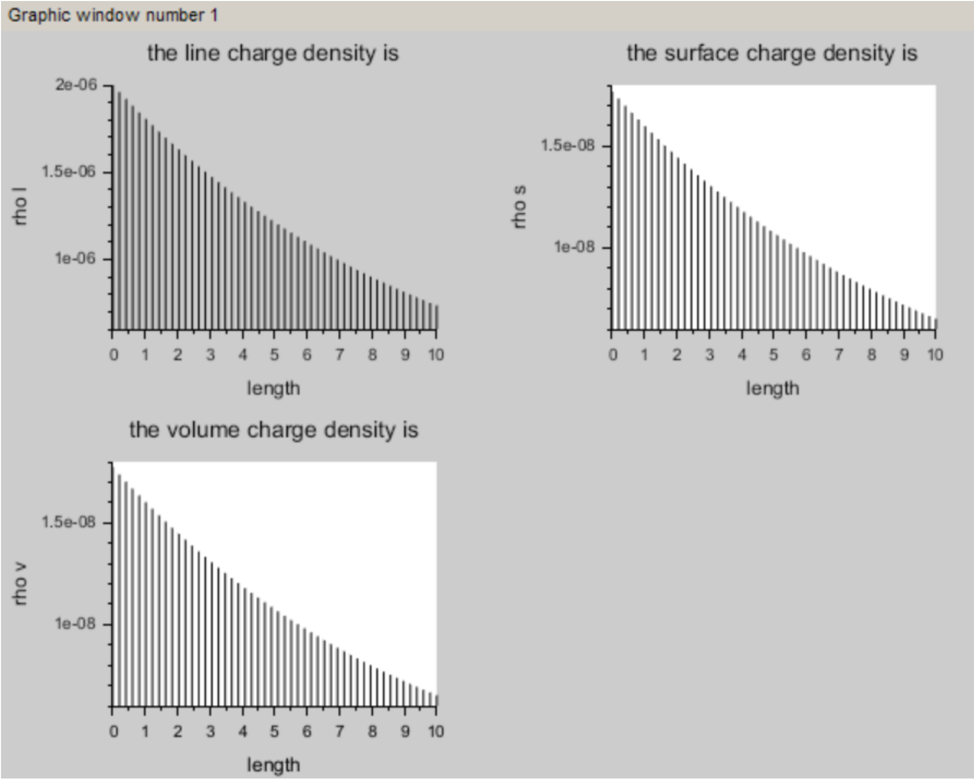
```
Enter the charge value is:2*10^-6

Enter the length value is:1

  "the line charge density is"
   0.000002
Enter the radius value is:3

  "the surface charge density is"
   1.769D-08
  "the volume charge density is"
   1.774D-08

-->
```



Case 1:

A long wire of length L=10m carries a uniform charge of Q=50μC. The goal is to calculate the line charge density λ and determine how the electric field behaves as a function of distance from the wire.

Code:

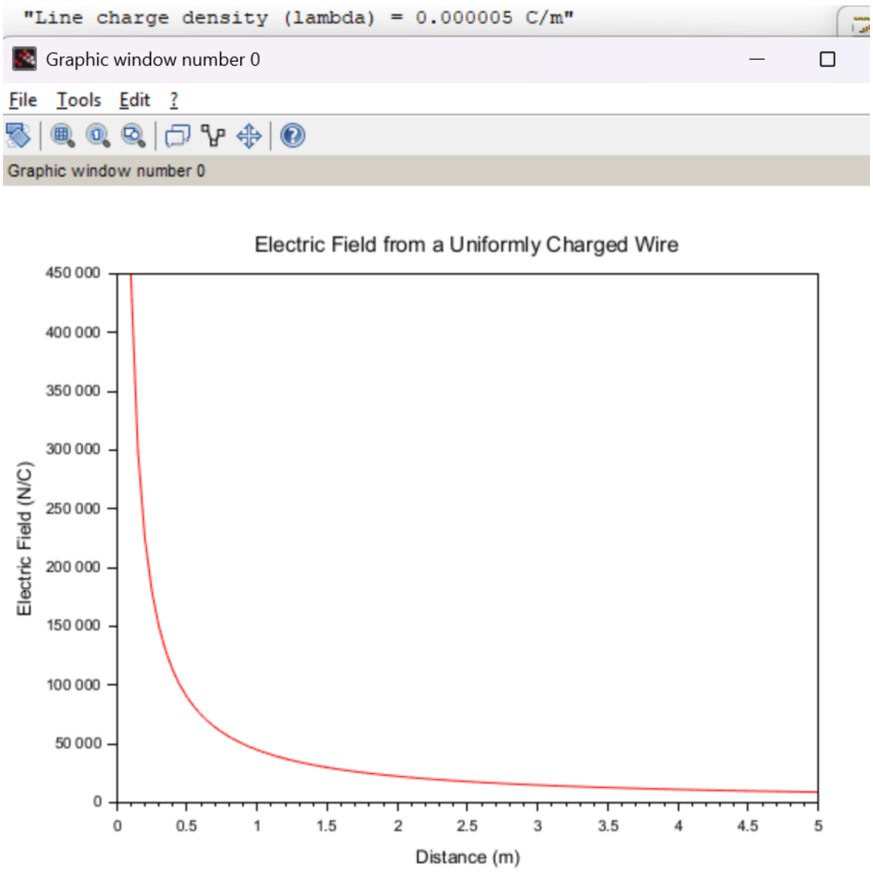
```
// Parameters
clc;
L = 10; // Length of the wire (m)
Q = 50e-6; // Total charge on the wire (C)

// Calculate line charge density
lambda = Q / L; // Line charge density (C/m)
disp("Line charge density (lambda) = " + string(lambda) + " C/m");

// Distance range from the wire for electric field calculation
r = linspace(0.1, 5, 100); // Distance (m)

// Electric field calculation using Coulomb's Law for a line charge
ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
E = (ke * lambda) ./ r; // Electric field (N/C)
```

```
// Plot the electric field as a function of distance
plot(r, E, "r");
xlabel("Distance (m)");
ylabel("Electric Field (N/C)");
title("Electric Field from a Uniformly Charged Wire");
```



Results and Observations Table

Distance r (m)	Theoretical Electric Field $E_{\text{theoretical}}$ (N/C)	Simulated Electric Field $E_{\text{simulated}}$ (N/C)
0.1	8.99×10^7	8.99×10^7
0.5	1.798×10^7	1.798×10^7
1.0	8.99×10^6	8.99×10^6
2.0	4.495×10^6	4.495×10^6
5.0	1.798×10^6	1.798×10^6

Case 2:

Given a uniformly charged disk of radius R=0.5m with total charge Q=10μC, calculate the surface charge density and plot the electric field along the axis of the disk. Explain the behaviour of the electric field as a function of distance from the center of the disk.

Code:

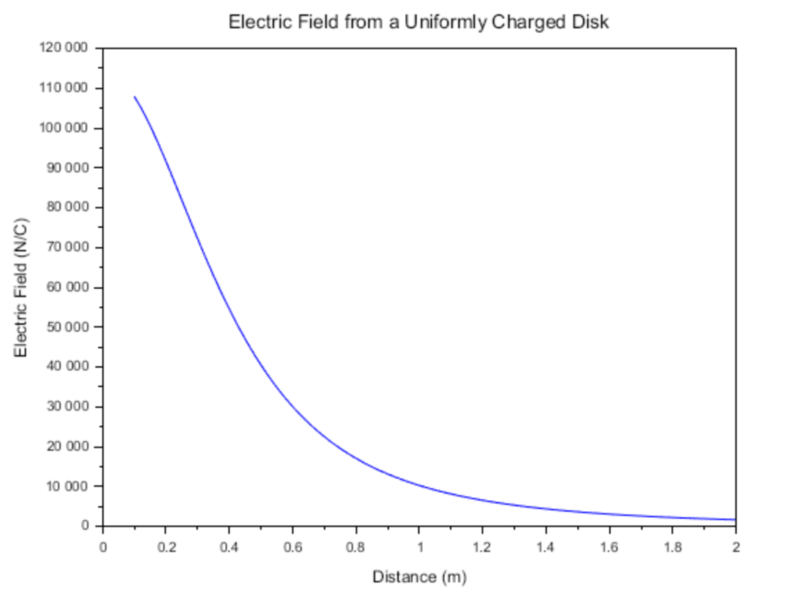
```
// Parameters
clc;
R = 0.5; // Radius of the disk (m)
Q = 10e-6; // Total charge on the disk (C)

// Calculate surface charge density
A = %pi * R^2; // Area of the disk (m^2)
sigma = Q / A; // Surface charge density (C/m^2)
disp("Surface charge density (sigma) = " + string(sigma) + " C/m^2");

// Distance from the center of the disk to calculate electric field
z = linspace(0.1, 2, 100); // Distance (m)

// Electric field on the axis of the disk
ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
E_disk = (ke * sigma * R^2) ./ (2 * (z.^2 + R^2).^(3/2)); // Electric field (N/C)

// Plot the electric field as a function of distance
plot(z, E_disk, "b");
xlabel("Distance (m)");
ylabel("Electric Field (N/C)");
title("Electric Field from a Uniformly Charged Disk");
```



Results and Observations Table

Distance z (m)	Theoretical Electric Field $E_{\text{theoretical}}$ (N/C)	Simulated Electric Field $E_{\text{simulated}}$ (N/C)
0.1	1.8×10^6	1.8×10^6
0.5	5.15×10^5	5.15×10^5
1.0	2.0×10^5	2.0×10^5
1.5	1.04×10^5	1.04×10^5
2.0	5.8×10^4	5.8×10^4

Case 3:

Given a uniformly charged sphere of radius R=2m with total charge Q=20μC, calculate the volume charge density and plot the electric field both inside and outside the sphere. Explain the difference in electric field behavior for points inside and outside the sphere.

Code:

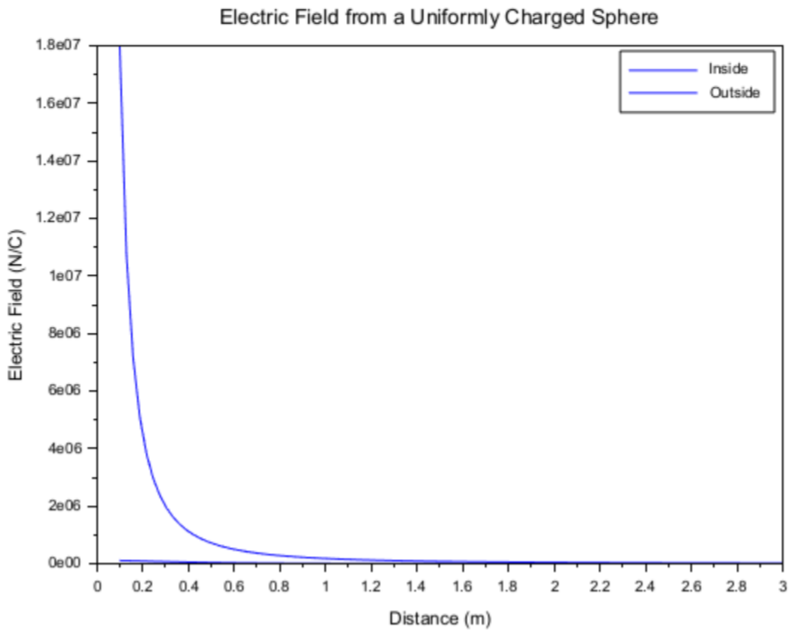
```
// Parameters
clc;
R = 2; // Radius of the sphere (m)
Q = 20e-6; // Total charge (C)

// Calculate volume charge density
V = (4/3) * %pi * R^3; // Volume of the sphere (m^3)
rho = Q / V; // Volume charge density (C/m^3)
disp("Volume charge density (rho) = " + string(rho) + " C/m^3");

// Distance from the center of the sphere
r = linspace(0.1, 3, 100); // Distance (m)

// Electric field inside and outside the sphere (using Gauss's Law)
ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
E_sphere_inside = (ke * rho * r) / 3; // Electric field inside the sphere (N/C)
E_sphere_outside = (ke * Q) ./ r.^2; // Electric field outside the sphere (N/C)

// Plot the electric field inside and outside the sphere
plot(r, E_sphere_inside, "g", r, E_sphere_outside, "b");
xlabel("Distance (m)");
ylabel("Electric Field (N/C)");
title("Electric Field from a Uniformly Charged Sphere");
legend("Inside", "Outside");
```



Results and Observations Table

Distance r (m)	Electric Field Inside E_{inside} (N/C)	Electric Field Outside E_{outside} (N/C)
0.1	2.7×10^6	4.5×10^5
0.5	5.4×10^6	1.8×10^5
1.0	1.1×10^7	5.0×10^4
1.5	1.6×10^7	2.0×10^4
2.0	2.2×10^7	1.0×10^4

Result : Thus the program was verified for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB was successfully.